

Malek Badawi

Baton Rouge, LA | (225) 931-2624 | malekbadawi21@gmail.com | malekbadawi.com
github.com/malekbadawi21 | linkedin.com/in/malekbadawi21

EDUCATION

Louisiana State University (LSU), Baton Rouge, LA

Expected May 2028

Bachelor of Science in Computer Science (Software Engineering) | **Minors:** Mathematics and Honors Research *GPA: 4.17/4.30*

Honors: Stamps Scholar (full-ride national scholarship), President's Future Leaders in Research, President's Honor Roll

Coursework: LLM Application Development (Fall 2026) | Advanced Data Structures and Algorithms, Discrete Structures, Numerical Methods, Computer Organization and Design, Differential Equations/Linear Algebra, Object-Oriented Programming

University College London – Artificial Intelligence Concepts & Applications, Summer 2025 (Distinction)

EXPERIENCE

Undergraduate Researcher

March 2026 – Present

Pennington Biomedical Research Center – Imaging Department

Baton Rouge, LA

- Cross-checked the lab's BRISQUE QC pipeline against manual review and showed it ranked motion-corrupted scans among the highest, due to natural-image statistics missing MRI artifacts, and drove the pivot to radiologist-validated MRIQC.
- Consolidated FreeSurfer cortical-surface outputs for approximately **460 subjects** into a unified symlink-based directory, automating version-conflict resolution across inconsistent naming conventions and eliminating case-by-case file selection.
- Deployed MRIQC in an Apptainer container on a shared Linux/NFS server and built a BIDS-compliant pilot dataset for 23 subjects, including DICOM-to-NIfTI conversion, symlink organization and programmatic validation with pybids.
- Developed Bash and MATLAB batch-processing workflows across seven concurrent studies containing **500+ subjects**, adding dry-run validation and failure safeguards while resolving file-processing and variable-scope errors.

Software & Data Engineering Intern

September 2025 – June 2026

Automotive Data Analytics

Remote

- Built and tested a three-endpoint SMS authentication service using JavaScript, Azure Logic Apps, Azure Table Storage and Twilio, supporting OTP requests, verification and session validation with five-minute code expiration and 24-hour sessions.
- Designed isolated Logic App workflows for each authentication endpoint and restructured token storage to enforce one active session per user after identifying that expired tokens were not being removed from Azure Table Storage.
- Built Microsoft Fabric ETL pipelines that consolidated **10,000+ customer records** from multiple dealership systems into a centralized lakehouse, automated recurring refreshes and supported Power BI reports that replaced manual performance-report compilation.

PROJECTS

All You Can Eat | *TypeScript, Next.js, React, SQLite, Vitest, GitHub Actions*

- Built a natural-language query parser and multi-objective search engine in TypeScript and Next.js, scoring **9,900 fast-food orders** per query across nutrition goals, calories, price, and menu constraints.
- Diagnosed that nutrition-only optimization produced unorderable results and designed a meal-grammar layer using category role, daypart, and portion weight; added diversified ranking and **24 regression tests** enforced through GitHub Actions.

Golf Swing Analyzer | *Python, OpenCV, MediaPipe*

- Developed a Python computer-vision pipeline using OpenCV and MediaPipe to extract body landmarks from swing videos and calculate hip rotation, shoulder rotation, spine angle and tempo on a frame-by-frame basis.

Personal Portfolio Site | *React, Vite, React Router, CSS, Vercel*

- Designed and built a single-page application portfolio with client-side routing and a custom design system; deployed with CI on push at malekbadawi.com.

SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, SQL, Bash, HTML/CSS, MATLAB

Frameworks & Libraries: React, Next.js, Node.js, Vitest, OpenCV, MediaPipe, NumPy, pandas

Tools & Platforms: Git/GitHub, GitHub Actions, Linux/Unix, REST APIs, Postman, SQLite, Azure, Microsoft Fabric, Power BI, Twilio, Vercel, Apptainer/Singularity

Research Computing: FreeSurfer, MRIQC, pybids, BIDS, NIfTI, DICOM

Spoken: Arabic (fluent), English (fluent), Spanish (conversational)

LEADERSHIP

Student Government Senator

August 2025 – Present

Louisiana State University – Baton Rouge, LA

- Elected by the College of Engineering; serve on the Student Auxiliary Services and Academic Affairs Committees, contributing to initiatives involving campus technology resources and ethical AI policy.